



EX 16.2.2

ASSUME THAT

- THE PROBABILITY OF RAIN TOMORROW IS 0.5 IF IT IS RAINY TODAY
- THE PROBABILITY OF BEING CLEAR IS 0.9 IF IT IS CLEAR TODAY

FORMULATE THE EVOLUTION OF THE WEATHER AS A MARKOV CHAIN (DEFINE STATES AND TRANSITION PROBABILITIES)

FIND THE STEADY STATE PROBABILITY

EX 16.5.1

REPLACE THE PROBABILITIES OF THE PREVIOUS EXERCISE WITH ARBITRARY VALUES i.e.

0.5 BECOMES $\alpha \in (0, 1)$

0.9 BECOMES $\beta \in (0, 1)$

FIND THE STEADY STATE PROBABILITY

EXERCISE

FIND THE STEADY STATE PROBABILITY AND
DRAW THE MARKOV CHAIN, GIVEN THE FOLLOWING
TRANSITION MATRIX

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

EXERCISE (EXAM - USE!)

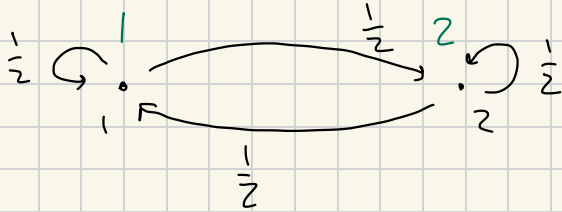
GIVEN THE FOLLOWING TRANSITION MATRIX

$$P = \begin{bmatrix} 0.5 & 0.4 & 0.1 & 0 & 0 \\ 0.8 & 0.1 & 0 & 0 & 0.1 \\ 0 & 0 & 0.6 & 0.6 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- DRAW THE MARKOV CHAIN
- WHAT IS THE PROBABILITY OF ENDING IN STATE 5 WHEN STARTING IN STATE 1?
- FOR INITIAL STATE 1, CALCULATE THE STEADY STATE DISTRIBUTION
- HOW MANY TIMES WE WILL BE IN STATE 1 BEFORE LEAVING THE TRANSITION STATES?

EXERCISE with TIME

IMAGINE WE DO NOT SPEND THE SAME
AMOUNT OF TIME IN EACH STATE



CALCULATE THE STEADY STATE DISTRIBUTION

EXERCISE with TIME 2

